

KORAUN S. HUMPHREY

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EDUCATION

North Carolina Agricultural and Technical State University

Greensboro, NC

B.S. Computer Engineering | Expected May 2029

GPA: 3.41/4.00 | Dean's List

Relevant Coursework: Digital Logic, Programming, Calculus I & II, Physics, Engineering Design

TECHNICAL SKILLS

Programming: C++/Arduino, Python, Java

Hardware / Embedded: DE10-Lite FPGA, Arduino, digital logic, microcontrollers, sensor integration

Engineering Tools: Quartus Prime, MATLAB/Simulink, Multisim, SolidWorks, AutoCAD

PROJECT EXPERIENCE

FPGA Synchronous Up/Down Counter | *Quartus Prime* | *DE10-Lite FPGA* | *Digital Logic*

- Designed and implemented a 2-bit synchronous up/down counter using a Moore finite-state machine, D flip-flops, next-state logic, and a 7-segment display.
- Developed state-transition logic and verified system behavior through waveform simulation before programming and testing the design on a DE10-Lite FPGA.

Arduino Ultrasonic Proximity Indicator | *Arduino* | *C++* | *Ultrasonic Sensor*

- Developed a proximity detection system that measures object distance in real time and uses C++ control logic to change LED indicators based on distance range.
- Integrated and tested microcontroller software, sensor input, and breadboard circuitry to verify reliable physical responses.

FBS Offensive Performance Analyzer | *Python* | *Data Processing* | *Algorithms*

- Developed a Python application to process 2023 FBS offensive statistics across multiple performance categories.
- Implemented functions, sorting, and ranking logic to compare team records and generate a composite Top 10 offensive ranking.

Room Presence Logger | *Personal Embedded Systems Project* | *In Progress*

- Developing a sensor-based embedded system to detect room occupancy and log presence data over time, emphasizing hardware-software integration and data collection.

WORK EXPERIENCE

Kelly Services

High Point, NC

Assembly / Manufacturing Team Member

Summer 2026

- Performed hands-on assembly in a fast-paced manufacturing environment while following production procedures and quality expectations.
- Maintained accuracy and consistency across 40+ hour workweeks while collaborating with team members to meet daily production requirements.

Bojangles

Kinston, NC

Team Member

Summer 2025, 2026

- Worked evening shifts across multiple stations while simultaneously maintaining a full-time daytime manufacturing position.

LEADERSHIP & HONORS

NSBE - Member | IEEE - Member | National Honor Society - President | Aggie Merit Scholarship